Weighted Population of Individuals and Households

A reproducible analysis of poverty.dta | Estimates expanded from 9,600 sampled households

7,185,103	28,740,504	4.000
WEIGHTED HOUSEHOLDS	WEIGHTED INDIVIDUALS	PERSONS PER HOUSEHOLD

Purpose and key result

The objective is to estimate the national populations represented by the household records. Summing the supplied expansion factors gives **7,185,102.7 households** and **28,740,503.5 individuals**. Rounded to countable units, these are 7,185,103 households and 28,740,504 individuals. The ratio of the two unrounded totals is 4.000 persons per household.

Variables used

Variable	Role in calculation
psu_number + hh_number	Composite identifier; verified as unique, confirming one record per household.
hhsize	Number of usual household members represented by a sampled household.
hhs_wt	Census-2021-adjusted household expansion weight; one row represents this many households.
ind_wt	Individual expansion weight; dataset label defines it as adjusted household weight x household size.

How the weights are calculated

Let w_h be **hhs_wt** for sampled household *h*, and let m_h be **hhsize**. The file supplies the person expansion weight as $w_h m_h$. This identity was checked for every record within a relative tolerance of 0.000001 to allow float-storage rounding.

Households = Σ_h w_h | Individuals = Σ_h (w_h x m_h) = Σ_h ind_wt

Methodology

The Stata program **expansion_total** creates an indicator equal to one and estimates its total using the requested weight as a probability weight. The point estimate is the sum of the expansion weights. The do-file first validates unique household IDs, complete inputs, positive weights, household size of at least one, and the construction of **ind_wt**. It then estimates both national totals and exports exact, unrounded CSV results and diagnostic breakdowns.

Breakdowns and Interpretation

Rounded estimates; national totals may differ by one unit from displayed row sums because rounding occurs after estimation.

Province profile

Province	Households	Individuals	Persons / HH
Koshi	1,216,891	4,675,310	3.84
Madhesh	1,347,877	6,486,147	4.81
Bagmati	1,622,908	5,869,481	3.62
Gandaki	695,063	2,396,728	3.45
Lumbini	1,354,403	5,447,377	4.02
Karnali	372,795	1,473,115	3.95
Sudurpaschim	575,165	2,392,344	4.16
National	**7,185,103**	**28,740,504**	**4.00**

Residence profile

Residence	Households	Individuals	Share of individuals
Kathmandu	845,255	3,091,107	10.8%
Other urban	4,178,550	16,875,469	58.7%
Rural	2,161,298	8,773,927	30.5%

Interpretation

Each sampled household represents many households with similar survey-design characteristics. Therefore, the unweighted sample count of 9,600 must not be read as the household population. Bagmati has the largest estimated household population (1.62 million), while Madhesh has the largest estimated individual population (6.49 million). This difference reflects household size: the implied average is about 4.81 persons in Madhesh versus 3.62 in Bagmati. Other urban areas account for 58.7% of represented individuals, rural areas 30.5%, and Kathmandu 10.8%.

These are expansion totals for the target population encoded by the provided weights. The household weight label states that it was adjusted to the final 2021 census household count. The file does not provide strata or full documentation of nonresponse and calibration procedures, so this analysis does not reconstruct the original weight-development process or report design-based standard errors.

Reproducibility and checks

Quality check	Result
Household identifier	psu_number + hh_number uniquely identifies all 9,600 records
Missing calculation inputs	None
Positive weights / hhsize >= 1	Passed for every record
ind_wt = hhs_wt x hhsize	Passed within stated float-rounding tolerance

Run from this folder with: `StataMP-64.exe /e do weighted_population.do`. Outputs are **weighted_population.log**, **weighted_population_results.csv**, and the province and residence CSV files. The log is the audit record; the do-file contains detailed line-level comments.